

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Electronic Engineering (VLSI Design and Technology) SEMESTER - 5 Winter 2025 ( Regular )**

**Course : Bachelor of Technology (Electronic Engineering (VLSI Design and Technology)) Branch : Engineering and Technology**

**Semester : SEMESTER - 5**

**Subject Code & Name: BTEEV502 - MICROCONTROLLER & EMBEDDED SYSTEM**

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**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
  2. Question No. 1 will be compulsory and include objective-type questions.
  3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
  4. Use of non-programmable scientific calculators is allowed.
  5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which architecture uses separate memory spaces for program instructions and data?  
a) Von Neumann b) Harvard c) CISC d) RISC
  - 2 The 8051 microcontroller is classified as which type of instruction set architecture?  
a) RISC b) CISC c) Both d) Neither
  - 3 How many pins does the standard 8051 microcontroller package have?  
a) 20 b) 40 c) 64 d) 32
  - 4 How many programmable I/O ports are available in the 8051 microcontroller?  
a) 2 b) 3 c) 4 d) 5
  - 5 Which register is used for holding transmit and receive data in serial communication of 8051?  
a) TMOD b) SBUF c) TCON d) PCON
  - 6 The main difference between standard C and Embedded C is the support for:  
a) Hardware access b) None c) No pointers d) Slower
  - 7 In LCD interfacing with 8051, the RS pin is used for:  
a) Register Select b) Read/Write c) Enable d) Data
  - 8 For controlling direction and speed of a DC Motor with 8051, we typically use:  
a) Driver IC b) Direct c) Timer d) Interrupt
  - 9 Key design challenges in embedded systems are:  
a) Power/Cost/Real-time b) Unlimited memory c) No reliability d) Flexibility
  - 10 In an RTOS, the basic unit of execution is commonly referred to as:  
a) Task/Thread b) Process c) Kernel d) Interrupt
  - 11 Common types of task scheduling used in RTOS include:  
a) Preemptive/Round-robin b) FIFO c) None d) Random
  - 12 Multitasking in an RTOS is achieved by:  
a) Unlimited b) Single c) Concurrent switching d) No threads

- Q2. Solve the following.
- A) Explain the criteria for selection of a microcontroller for an embedded application. Discuss at least five key parameters with suitable examples. 6
  - B) Draw the basic block diagram of the 8051 microcontroller and explain the function of each major block. 6
- Q3. Solve the following.
- A) Describe the interrupt structure of 8051. List all interrupt sources, their default priority, vector addresses, and explain how to enable/disable individual interrupts. 6
  - B) Write an Embedded C program to count the number of external pulses applied on INT0 pin using interrupt. Display the count on Port 2 continuously (use 8-bit binary display). 6
- Q4. Solve Any Two of the following.
- A) Write an Embedded C program to interface a DAC0808 with 8051 to generate a triangular waveform (0V to 5V ramp up and down continuously) using Port 0 as data lines. 6
  - B) Write an Embedded C program to interface a 16×2 LCD in 8-bit mode. Display "CITY: ABC" on first line and "TEMPERATURE: 35°C" on second line with continuous updating of temperature value. 6
  - C) Write an Embedded C program to control a stepper motor interfaced with 8051 using ULN2003 driver. Rotate the motor in clockwise direction at 60 RPM for 10 revolutions, then change direction to anti-clockwise. 6
- Q5. Solve Any Two of the following.
- A) Classify embedded systems based on complexity and performance. Provide two practical examples for each category. 6
  - B) Describe the various design challenges in embedded system development. Suggest solutions for these challenges. 6
  - C) Explain the basic concepts of operating system services required in embedded applications: task management, inter-task communication, synchronization, and memory management. 6
- Q6. Solve Any Two of the following.
- A) Differentiate between multiprocessing, multitasking, and multithreading in the context of RTOS. Explain with suitable diagrams. 6
  - B) Describe the differences between hard real-time and soft real-time systems. Give two examples of applications for each type. 6
  - C) What is context switching in RTOS? Explain the steps involved in context switching when a higher priority task becomes ready during execution of a lower priority task. 6

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